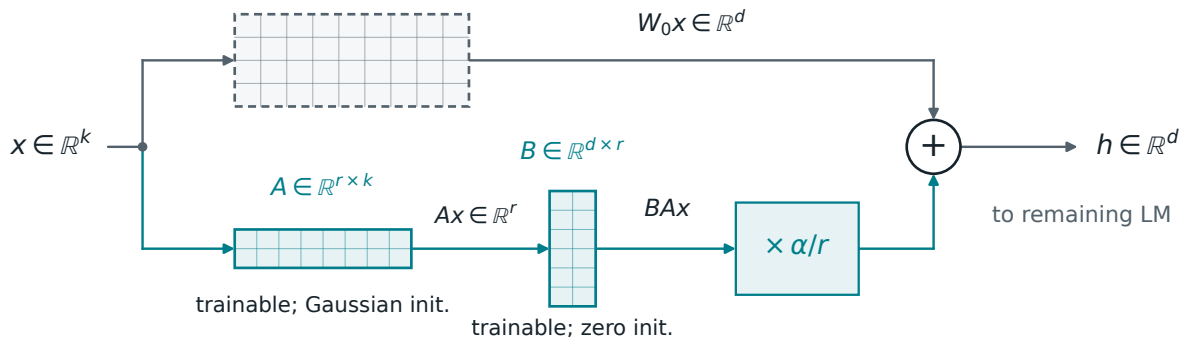


LoRA computation: train a low-rank branch, then merge

One adapted projection; repeated Transformer computation is omitted.

a Training

$W_0 \in \mathbb{R}^{d \times k}$ • frozen



Task loss updates A and B ; W_0 remains frozen.

$$h = W_0 x + s B (A x), \quad s = \alpha / r, \quad r \ll \min(d, k)$$

b Merged inference



Merge once for a fixed task; inference uses a single dense projection. No separate LoRA branch remains.